

A Project report on

PHOTOVOLTAIC MAXIMUM POWER- POINT TRACKING USING MODIFIED PARTICLE SWARM OPTIMIZATION UNDER PARTIAL SHADING CONDITIONS

In partial fulfilment of the requirements for the award of the degree of

Bachelor of Technology

in

Electrical and Electronics Engineering

Submitted by

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DEPARTMENT OF ELECTRICAL ENGINEERING
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APPROVAL SHEET

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MODIFIED PARTICLE SWARM OPTIMIZATION UNDER PARTIAL
SHADING CONDITIONS”

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DECLARATION

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LIST OF ABBREVIATIONS

PV	Photovoltaic
MPPT	Maximum power point technique
MPP	Maximum power point
GMPP	Global Maximum power point
LMPP	Local Maximum power point
UIC	Uniform irradiance condition
PSC	Partial shading condition
P-V	Power vs Voltage
I-V	Current vs Voltage
P&O	Perturb and observation
PSO	Particle swarm optimization
MPSO	Modified Particle swarm optimization
ANN	Artificial neural networks
FLC	Full load current
AI	Artificial intelligence
DC	Direct current

Abstract

This project report discusses the performance of a modified particle swarm optimization (MPSO) based maximum power-point tracking (MPPT) technique for the purpose of enhancing the efficiency of photovoltaic (PV) based power systems. Despite its widespread adoption in electric grids, the efficiency of a PV cell is still very less, which makes the transfer of maximum power from the source to loads of utmost importance. Moreover, the non-linear relationship existing between the input and output quantities of PV modules leads to suboptimum power transfer which is overcome by the use of MPPT controllers. There are a number of algorithms that exist to track the maximum power point (MPP), some of the most popular being the perturb and observe (P&O), and incremental conductance, also termed as the conventional techniques. These algorithms accurately track the MPP during uniform solar radiation conditions, but in practical scenarios such cases almost never occur, and the casting of shadows due to neighbouring structures/objects or atmospheric conditions can cause the partial shading of the PV array configuration. Shading complicates the MPP extraction as it generates multiple local MPPs with one global MPP. Conventional techniques give poor performance under these conditions, which require the development of intelligent techniques that enable an efficient extraction of the global maximum regardless of the surrounding environment.

The proposed method is evaluated for a number of shading profiles given to a series connection of four PV modules. Its performance is compared with that of the typical perturb and observe technique as well. The implementation, and theory of both the techniques are discussed. The evaluation is carried out in MATLAB/Simulink which uses a simple stand-alone PV system connected to a resistive load, the report also describes the modelling and theory behind the components involved in the PV system.

The report findings conclude that the MPSO method is always able to correctly track the GMPP, with a high efficiency percentage and stability, for all the radiation profiles it is tested for as compared to the P&O method.

Chapter 1

Introduction

1.1 Introduction

Photovoltaic technology is being increasingly adopted in today's power generation sector due to its advantages of cost-effectiveness, cleanliness, minimal carbon footprint, and abundance as compared to conventional energy sources. There exists a complex non-linear relationship between the output active power of the PV panel and its input parameters solar irradiance, and operating cell temperature which means that for any given load, there will be an unsatisfactory power extraction.

To overcome this challenge, maximum power point tracking techniques (MPPT) are developed to improve the PV system's overall efficiency and ensure that the operation point is always at the maximum power point (MPP).

For a given configuration of solar modules, under uniform irradiance conditions (all modules receive equal irradiance) conventional strategies such as perturb and observe (P&O), and incremental conductance can easily track the peak power of the system. However, practical implementations of PV systems find it challenging to install panels where there are no partial shading conditions due to the interference of buildings, towers, trees, clouds, etc. This interference causes the power-voltage characteristic curve to generate multiple peaks, including one global MPP, and many other local peaks, as shown in figures 1.1 and 1.2. PSC introduces complexity in extracting the MPP of the system.

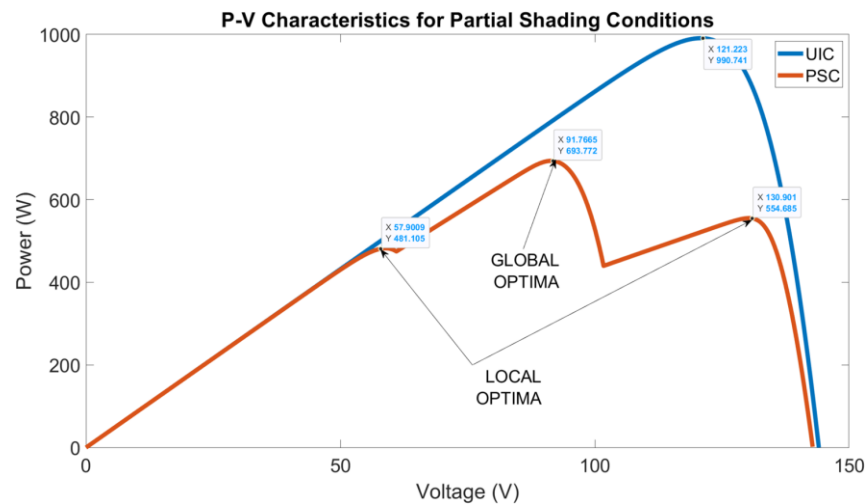


Figure 1.1. P-V Curves for UIC and PSC for a series of four PV arrays

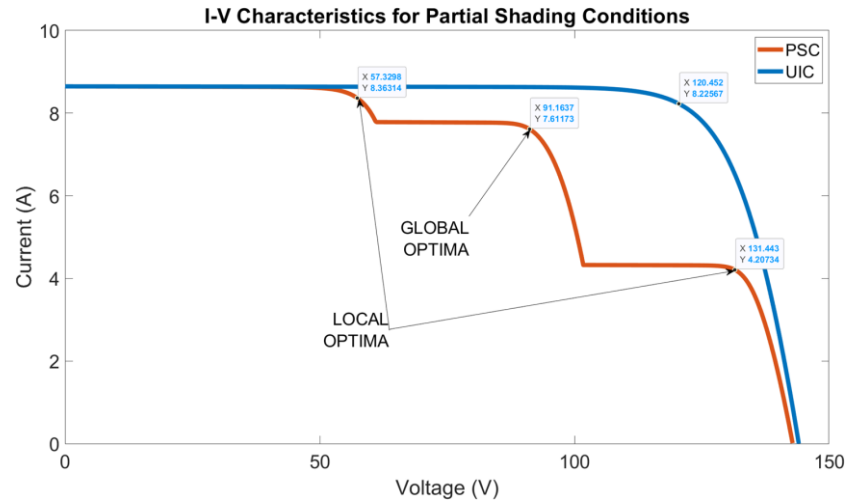


Figure 1.2. I-V Curves for UIC and PSC for a series of four PV arrays

1.2 Motivation

The conventional techniques fail to track the global MPPs in the case of PSC. One area of focus is to improve the performance of PV arrays is to develop better MPPT techniques that have the following features:

- A. It accurately and reliably tracks the GMPP, regardless of the irradiance condition, uniform or not.
- B. It possesses a fast convergence time.
- C. It is simple and easy to implement.
- D. It is cost-effective.

1.3 Objectives of Report

The main contributions of this report are as follows:

- A. Implement a modified particle swarm optimization MPPT always tracks the GMPP with a high degree of preciseness to enhance the efficiency of PV systems under PSC.
- B. MPSO performance is compared with a P&O conventional technique to extract the true maximum power from the PV system at different cases of PSCs.

1.4 Organisation of Report

- A. Chapter 2: This chapter discusses the various techniques that exist in literature regarding MPPT algorithms, it briefly describes the principle behind each method and lists out its advantages and disadvantages.
- B. Chapter 3: This chapter deals with the modelling of the elements used for the purpose of testing out the MPPT algorithm. It details the principle behind their operation and gives the relevant equations that describe their behaviour.
- C. Chapter 4: This discusses the two MPPT techniques used in this report. It describes the methods, gives the flowchart required for its implementation, and all the relevant equations involved in its working.
- D. Chapter 5: This chapter lists out the MATLAB/Simulink models of the system involved, all the relevant system parameters required for simulation, and the input cases used to evaluate the performance of the algorithms.
- E. Chapter 6: This chapter shows the power and voltage waveforms tracked by both the algorithms; it also discusses its implications.
- F. Chapter 7: This chapter presents the conclusions and also the improvements that can be made for any future developments.

Chapter 2

Literature Study

In literature, there have been a number of algorithms developed for searching the global MPP under conditions of partial shading. In general, MPPT methods can be broadly classified into three groups:

- A. Conventional MPPT
- B. AI-based MPPT
- C. Hybrid MPPT

2.1 Conventional MPPT

2.1.1 Perturb and Observe (P&O)

This involves introducing a small disturbance in the duty ratio of the power convertor, or in the operating voltage of the PV array. It measures the PV array voltage and current, and depending on the change in power, the subsequent perturbations are given.

The process is repeated periodically until MPP is reached and it oscillates around the MPP. This oscillation can be minimized by reducing the step size, with a compromise in convergence time. Refer to Section 4.1 for its implementation.

2.1.2 Incremental Conductance

This method is based on the fact that the slope of the PV array power curve is zero at the MPP, positive on the left of the MPP, and negative on the right.

Since,

$$\frac{dP}{dV} = \frac{d(IV)}{dV} = I + V \frac{dI}{dV} = I + V \frac{\Delta I}{\Delta V} \quad (2.1)$$

Then,

$$\left\{ \begin{array}{l} \frac{\Delta I}{\Delta V} = -\frac{I}{V}, \quad \text{at MPP} \\ \frac{\Delta I}{\Delta V} > -\frac{I}{V}, \quad \text{left of MPP} \\ \frac{\Delta I}{\Delta V} < -\frac{I}{V}, \quad \text{right of MPP} \end{array} \right. \quad (2.2)$$

The MPP can be tracked by comparing the instantaneous conductance (I/V) to the incremental conductance ($\Delta I/\Delta V$). Once the MPP is reached, the operation is maintained at this point unless any change in ΔI is noted.

2.1.3 Disadvantages

- A. Lack of adaptive, self-learning step-size capabilities.
- B. High steady-state error, power oscillation at MPP, and slow transient response.
- C. Inability to find GMPP, trapping at local MPP and incorrect perturbation direction under PSC.

2.2 AI-based MPPT

2.2.1 Artificial Neural Networks

This is an architecture that consists of a network of interconnected nodes that predicts an output from an input set of experimentally obtained data. The accuracy of predictions improves with more ‘training’, which refers to giving the neural network a set of inputs along with its associated outputs to establish a linear relationship between them.

When applied to PV systems, the inputs are PV Voltage, PV Current, Irradiance, and Temperature. The output can be the Duty Cycle at GMPP.

2.2.1.1 Advantages

- A. Fast response and convergence speed.
- B. Slight fluctuation in steady-state.

2.2.1.2 Disadvantages

- A. A massive dataset is required, preparing which can be complex and time-consuming
- B. Periodic tuning is required due to environmental changes and aging.
- C. Accuracy of the model is highly system dependent, which means that if in an existing array configuration, another array is added, then the entire network will have to be trained again.

2.2.2 Swarm Intelligence Methods

This is a family of evolutionary optimization algorithms inspired by the collective behaviour of social animals, such as ants, bees, birds, wolves, etc. These techniques use the concept of 'swarm' to mimic the cooperation and competition of these animals.

These methods involve a population of randomly generated particles (represent possible solutions) that interact with each other and with the environment to collectively search for the optimal solution.

Some methods used for MPPT applications are:

A. Particle Swarm Optimization

A swarm of particles moves through the search space, with each particle updating its position based on its own best-known position, and the global best-known position of the swarm.

Refer to Section 4.2 for its implementation.

B. Grey Wolf Optimization

This is inspired by the social behaviour of grey wolves, in which a population of virtual wolves mimics the hunting and social hierarchy of real wolves to find the optimal solution.

The population updates its locations based on their distance from the three best-performing wolves of the pack, referred to as the α , β , and δ wolves.

C. Firefly Algorithm

This is inspired by the flashing behaviour of fireflies, which use bioluminescence to attract mates or prey. It involves the swarm moving through the solution space, with its brightness determined by the value of the optimizing function.

These are the descriptions of some of the many swarm intelligence-based techniques that are present in literature used for MPPT implementation.

2.2.2.1 Advantages

- A. No requirement of a dataset.
- B. High ability in searching for the GMPP regardless of its location.
- C. High tracking accuracy.
- D. Simple structure, easy implementation.

2.2.2.2 Disadvantages

- A. Oscillations because of random search.
- B. Large computational burden.
- C. Algorithm parameters need to be carefully set.

2.3 Hybrid MPPT

These techniques involve the integration of two or more MPPTs from both AI and conventional techniques. It combines the advantages of both, whilst overcoming the disadvantages of the individual techniques.

Some implementations of Hybrid MPPT are as follows:

- A. ANN with P&O algorithm
- B. Adaptive Neuro-Fuzzy Inference System
It has the advantages of both ANN and Fuzzy Logic Controllers. The ANN is trained to estimate the optimal MPP and used to drive an FLC-based MPPT.
- C. PSO with IC algorithm

2.3.1 Advantages

- A. Fast convergence speed, and high accuracy

2.3.2 Disadvantages

- A. Relatively complex.
- B. Longer computational time.
- C. Costly to implement.

Chapter 3

Modelling of a PV System

3.1 PV Arrays

PV cells are the basic elements of solar power generation, that are capable of converting the energy contained in photons of light into an electrical voltage and current. The output from a single cell is small, around 0.5V. However, most applications use a module that is a series and/or parallel connection of pre-wired cells encased in tough, weather-resistant packages. A module can have 36, 60, or even 72 cells. Multiple modules can be connected in series or parallel to obtain the desired power required.

The output features of PV arrays are non-linear and time-varying and are easily influenced by the incident solar radiance, cell temperature, and load conditions resulting in a solar energy utilization that cannot always be maximized. Figures 3.1 and 3.2 show the variation in I-V and P-V characteristics of a module with changing temperature. Figures 3.3 and 3.4 show the variation in characteristic curves for changing solar irradiation. In both cases, it can be observed that the position of MPP changes with varying cell temperatures and irradiances.

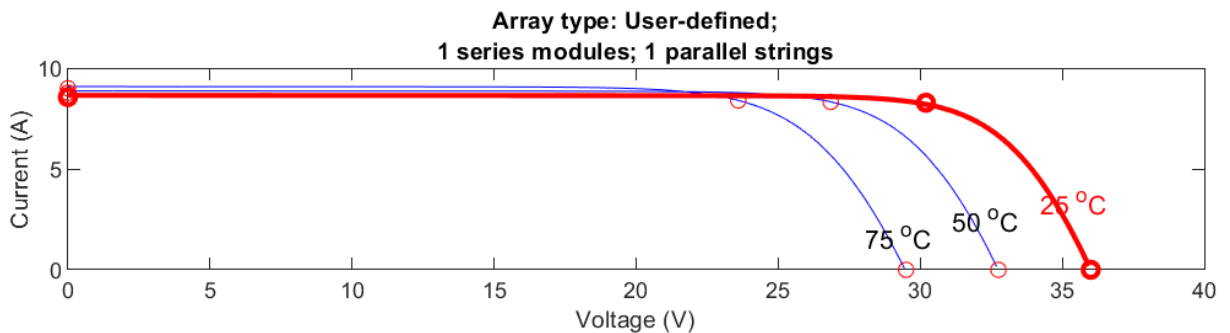


Figure 3.1 I-V characteristic curves for temperatures of 25°C, 50°C, 75°C at 1 kW/m²

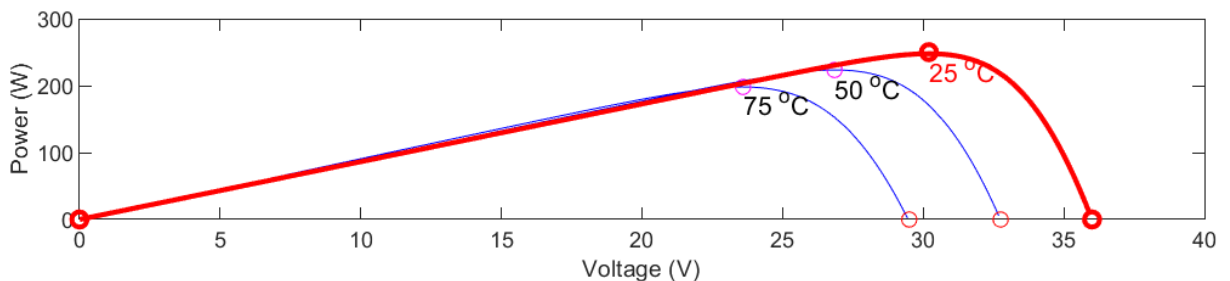


Figure 3.2 P-V characteristic curves for temperatures of 25°C, 50°C, 75°C at 1 kW/m²

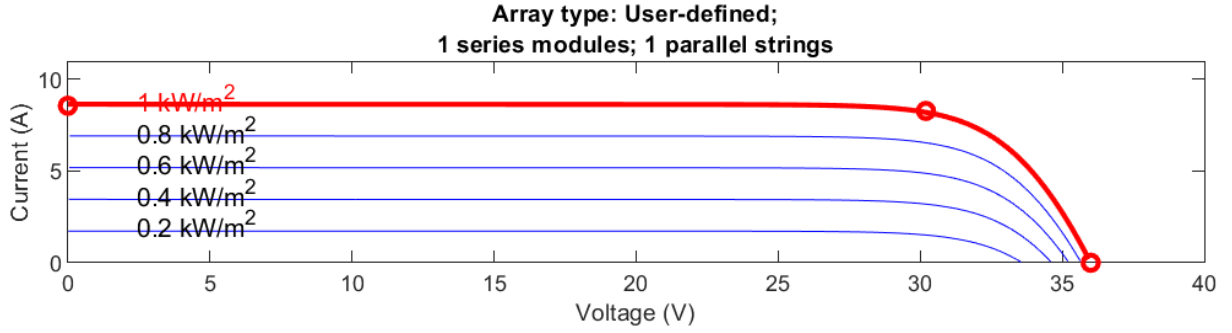


Figure 3.3 I-V characteristic curves for irradiances of 1 kW/m², 800 W/m², 600 kW/m², 400 W/m², 200 W/m² at 25°C

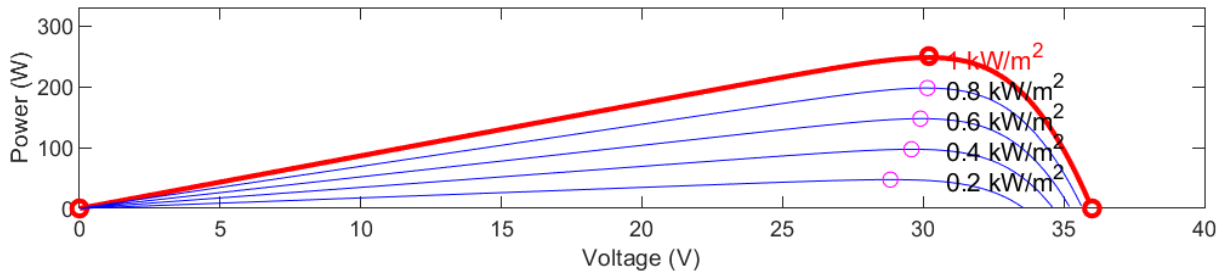


Figure 3.4 P-V characteristic curves for irradiances of 1 kW/m², 800 W/m², 600 kW/m², 400 W/m², 200 W/m² at 25°C

PV cells are made up of semiconductor material PN junctions that generate a direct voltage when receiving solar radiation. When modelling them, they are represented as a parallel connection of a constant current source and a forward diode. The equivalent circuit is shown in figure 3.5.

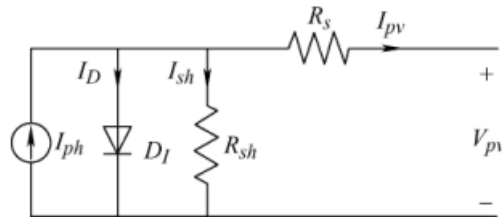


Fig. 3.5 Equivalent circuit of PV cell

In the circuit, I_{ph} is the photon-provided current, and its value depends on the area of cell that receives irradiance, and surround temperature; I_D is the reverse current in PV cell; I_{sh} is the shunt current that goes through the bypass resistor R_{sh} , that is generated owing to the battery edge and metal bridge leakage on the metal electrode; R_s is the series resistor that represents the resistivity of material and contact resistance; I_{pv} and V_{pv} are the solar PV current and voltage respectively.

The PV cell is simulated in MATLAB/Simulink using the 'PV Array' block in the Simscape library, which models the following equation:

$$I_{pv} = N_p I_{ph} - N_p I_o e^{\frac{q(V_{pv} + I_{pv} R_s)}{N_s A K T} - 1} - \frac{V_{pv} + I_{pv} R_s}{R_{sh}} \quad (3.1)$$

where, q is the electron charge, 1.6×10^{-19} C, N_s is the number of cells in series, N_p is the number of cells in parallel, I_o is the reverse saturation current under standard test conditions, T is cell temperature, A is the diode ideality constant, and K is the Boltzmann constant, 1.38×10^{-23} J/K.

3.2 Bypass Diodes

The output of a PV module can be reduced drastically, even if a small portion of it is shaded. The effect of a single shaded cell in a long string of cells can drop the output power by more than half. External diodes, referred to as bypass diodes are connected in parallel with the modules to mitigate the impacts of shading. Figure 3.6 shows a string of cells with one shaded cell.

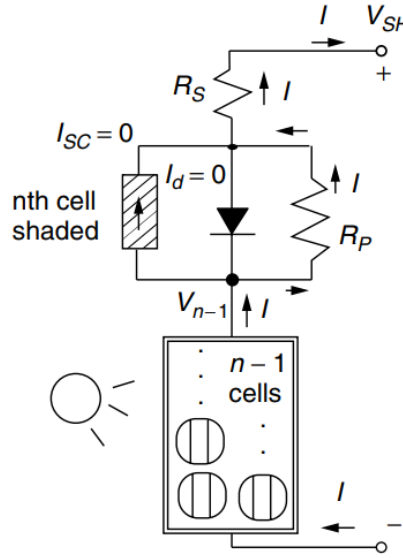


Fig. 3.6 A module with N cells, in which the top cell is shaded.

With the top cell shaded in Fig. 3.6, the current source I_{ph} has been reduced to zero. This results in the diode being reverse biased, which means that the entire current of the module flows through R_s and R_p of the shaded cell. The top cell, rather than adding to the output voltage, actually reduces it. Moreover, the heat dissipated in the resistances creates hot-spots, and if it exceeds a certain limit then it can permanently damage the cell.

The voltage drop is corrected using the bypass diodes, as shown in figure 3.7. Under normal conditions, the diode will be reverse biased. But shading across the cell causes the diode to

forward bias limiting the drop across the cell to $0.6 - 0.7\text{V}$, instead of the large drop that occurs in its absence.

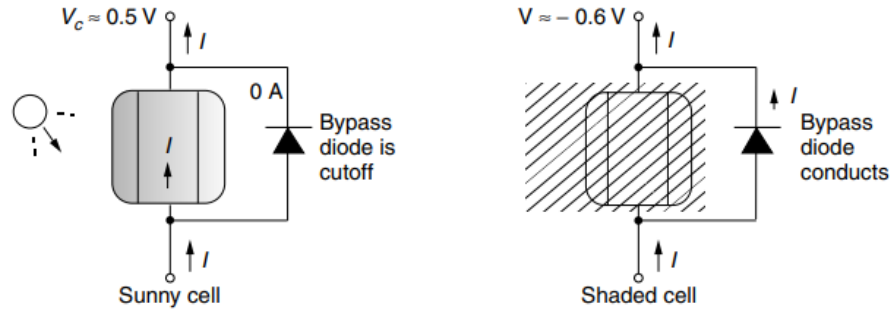


Fig. 3.7 Mitigation of shading with bypass diodes

The output characteristics of the modules also change with the presence of bypass diodes, as shown in Figures 3.8 and 3.9. There are multiple local peaks, with only one global peak.

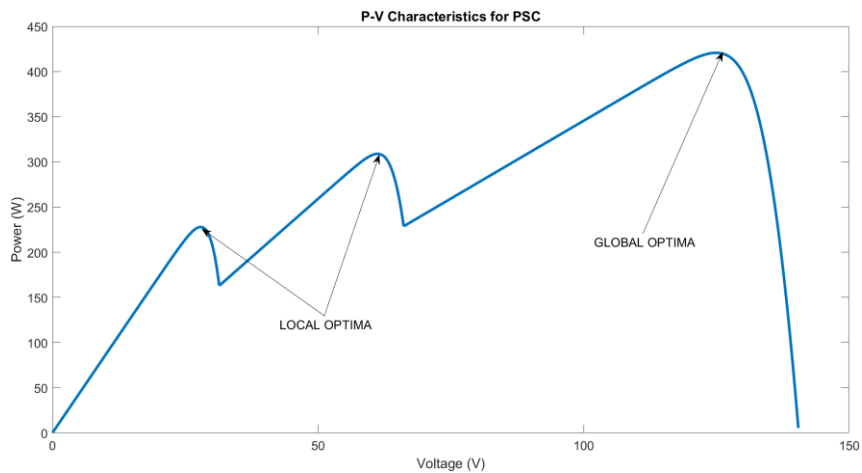


Fig. 3.8 P-V characteristics under PSC

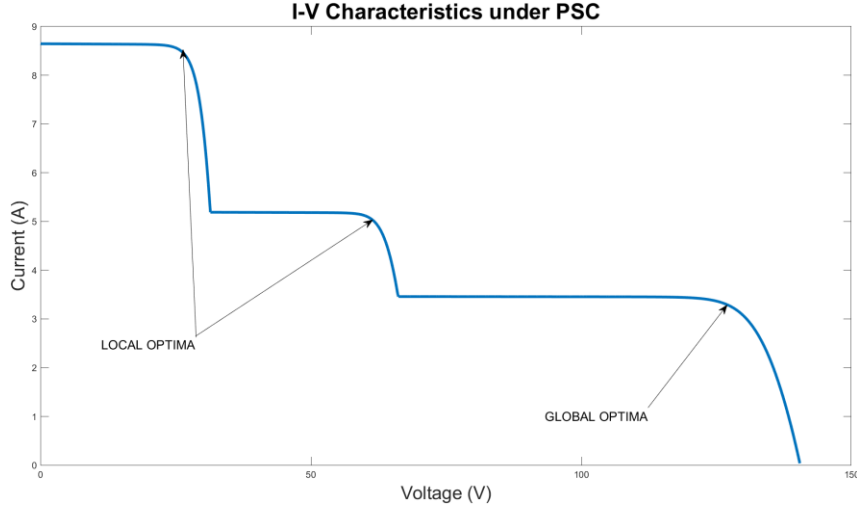


Fig. 3.9 I-V characteristics under PSC

3.3 DC-DC Boost Converter

DC-DC converters are extensively implemented in PV systems as a linking device between the PV modules and the consumers. They are used to match the process between the voltage of PV panels at the MPP and the load voltage for constant voltage applications.

The DC-DC converter used for this report is a boost convertor, that steps up the PV voltage level to a high output voltage level as per the following equation:

$$V_o = \frac{1}{1-D} V_i \quad (3.2)$$

The duty cycle varies between 0 and 1, meaning that the voltage ratio is always greater than 1, which causes the converter to be used as a step-up convertor. The circuit diagram of the boost convertor is shown in figure 3.10.

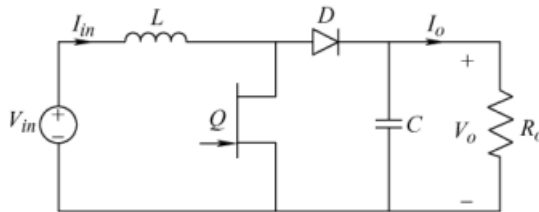


Fig. 3.10 Equivalent circuit of boost convertor

3.4 MPPT Controller

When the PV module is directly connected to a simple resistive load, the operating point will be the intersection of its characteristic curves with the load line having a slope of $\frac{1}{R}$, as shown in figure 3.11. With changing environmental conditions, the operating point slips causing the module to become less efficient. The solid maximum power point dots show the operating points that would result in maximum PV efficiency. The resistance value at MPP is given by,

$$R_{mpp} = \frac{V_{mpp}}{I_{mpp}} \quad (3.3)$$

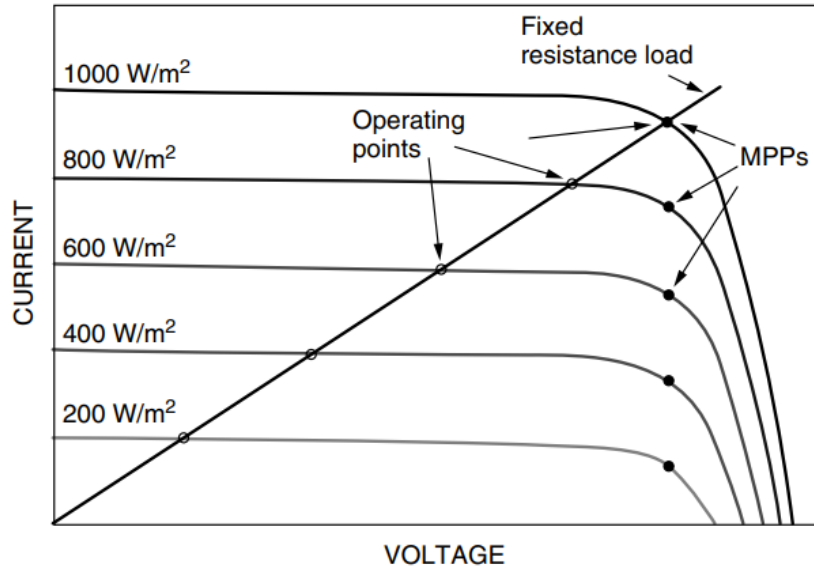


Fig. 3.11 Efficiency of a PV module connected to a fixed resistance will decline with changing irradiation.

To overcome this problem, an MPPT controller is used to keep the PVs operation at their highest efficiency points at all time. The maximum operation point is maintained by adjusting the duty cycle of the boost convertor. The duty cycle controls resistance seen by the PV modules such that it is equal R_{mpp} for the any given irradiance and cell temperature. The expression is given by,

$$D = 1 - \sqrt{\frac{R_{mpp}}{R_{load}}} \quad (3.4)$$

The overall block diagram of the PV system is given by figure 3.12.

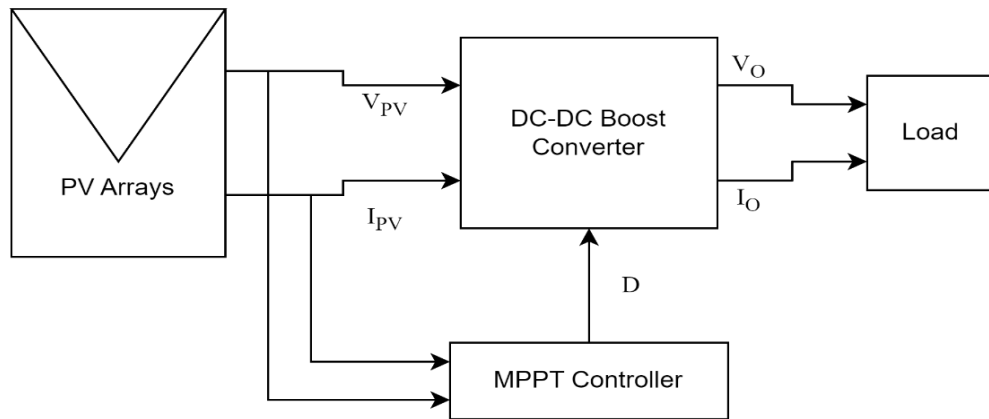


Fig. 3.12 Simple block diagram of PV System

Chapter 4

Methodology

There are two methods that are used for this study:

- A. Perturb and Observe (P&O)
- B. Modified Particle Swarm Optimization (MPSO)

4.1 Perturb and Observe

The P&O technique is one of the most widely used methods, especially for low-cost implementations. This measures the voltage and current of the PV array and depending on the change in PV power, introduces perturbations in the duty cycle of the convertor. Figure 4.1 shows the flowchart for this algorithm.

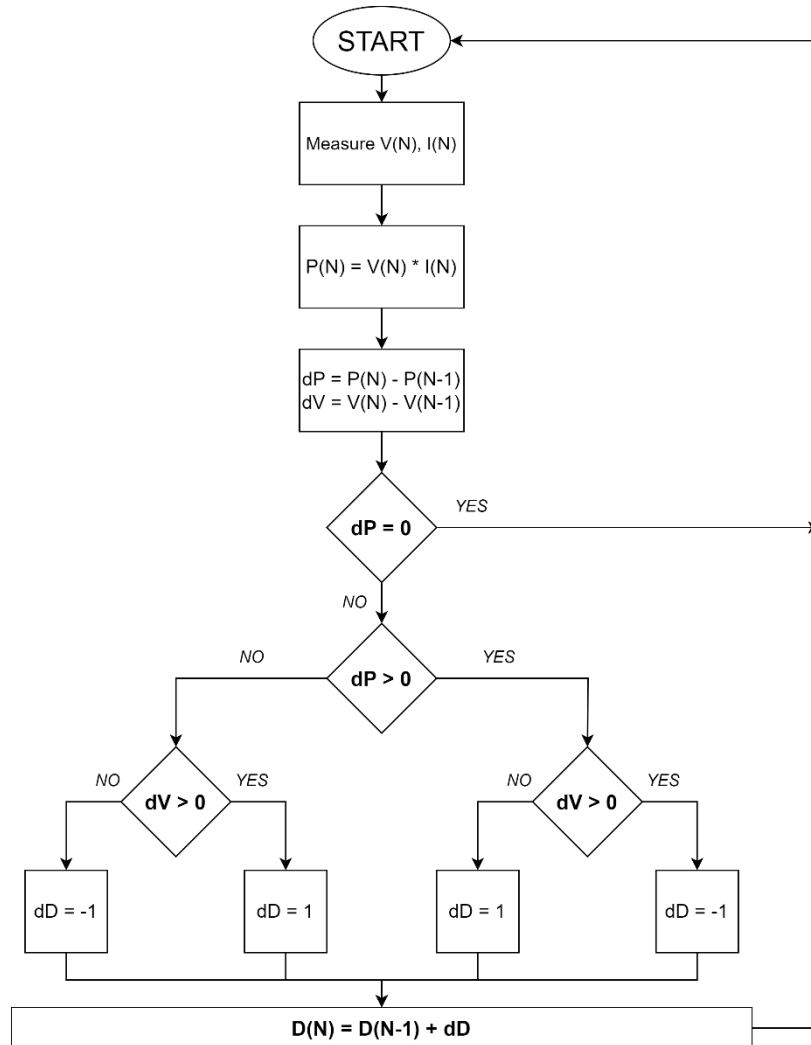


Fig. 4.1 Flowchart of P&O MPPT

The algorithm compares the value of power and voltage measured in the previous iteration with the values measured in the current iteration to decide the perturbation direction. Since for a boost convertor, the output voltage is inversely related with the duty cycle, the voltage is increased by decreasing the Duty Cycle and vice-versa.

The direction of perturbation will be decided as follows:

- A. Increase in Power; Increase in Voltage: Decrease Duty Cycle (Left of MPP)
- B. Increase in Power, Decrease in Voltage: Increase Duty Cycle (Right of MPP)
- C. Decrease in Power, Increase in Voltage: Increase Duty Cycle (Right of MPP)
- D. Decrease in Power, Decrease in Voltage: Decrease Duty Cycle (Left of MPP)

The above cases are summarised in Figure 4.2.

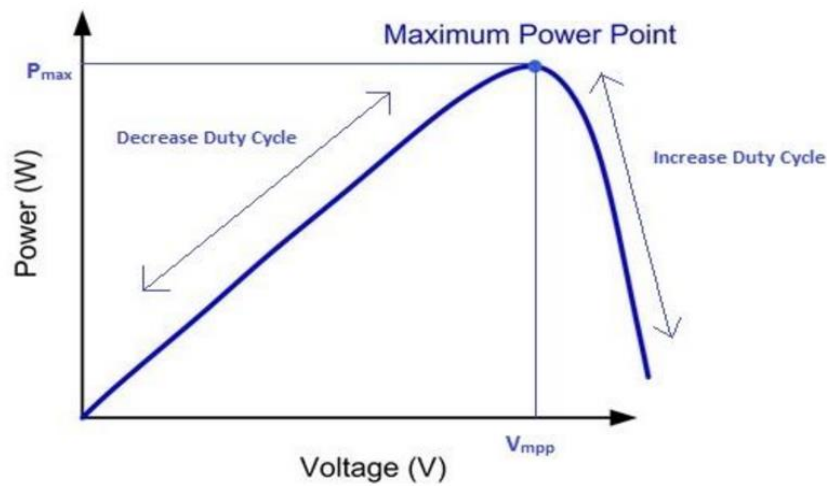


Fig. 4.2 Graphical representation of P&O method.

The process is repeated periodically until the MPP is reached. The system then oscillates about the MPP. The oscillations can be minimized by reducing the step size, at the expense of a slower convergence time.

A major disadvantage of this problem as will be shown in this report is its inability to track the GMPP under PSC and will settle at the first LMPP it encounters. Moreover, the oscillations around the MPP leads to loss in power.

4.2 Particle Swarm Optimization

Particle swarm optimization is an iterative, evolutionary optimization algorithm, that utilizes a swarm of particles (representing possible solutions) crawling around the search space. The particles move based on their velocities. The velocity of a particle is dependent on three components:

- A. **Inertia, ω** : It calculates the velocity and controls the effect of the previous velocity on the succeeding velocity. It determines whether the search happens globally or locally.
- B. **Individual Best, P_{best}** : As the particle, moves during execution, the objective function at each position is evaluated. The position where the particle attains its best objective function value is the individual best. It is updated every iteration.
- C. **Global Best, G_{best}** : The best position out of all the individual best positions is the global best. It represents the optimal solution to the objective function, in case of non-convergence of particles at the end of the search process.

The particles are generated randomly within the search space, and each particle is a candidate for the optimal solution. They are spread in the search space, with each moving the search space depending on its velocity. At each iteration, the new position is calculated for each particle from its velocity. The process continues till the stopping criteria is met, that is, after the iterations reach its maximum limit.

When using this method for differentiating between the GMPP and LMPP under PSC, the following considerations are made:

- A. Particle Position, X_i : Duty Cycle of the boost converter.
- B. Velocity, V_i : Variation in Duty Cycle
- C. Objective Function, to be maximized: PV Output Power

The equations that are used to decide the direction of the individual particles are as follows:

$$V_i^{t+1} = \omega V_i^t + c_1 r_1 (P_{best} - X_i^t) + c_2 r_2 (G_{best} - X_i^t) \quad (4.1)$$

$$X_i^{t+1} = X_i^t + V_i^{t+1} \quad (4.2)$$

where, ω is the inertia factor, c_1 and c_2 are two constant values, r_1 and r_2 are two random values between 0 and 1, i is the current particle and t is the current iteration.

The advantages of this method are that, it is computationally more efficient as it does not require to continuously compare the current positions with the previously obtained positions, instead only the global best and personal best of the individual particles is to be monitored.

On implementing this method for MPPT control, the following disadvantages are there:

- A. Large convergence time to find GMPP
- B. Possibility of the population to be concentrated in a local area within the search space, making the process to converge to the LMPP.
- C. Large power oscillations during the search process

To overcome the limitations of the conventional PSO method, a modified PSO algorithm is proposed in the next section.

4.2.1 Proposed Modified Particle Swarm Optimization

The modifications introduced in the calculations of velocity and position are as follows:

4.2.1.1 Velocity Step Function

Conventional PSO employs the step function V_i , which determined for i^{th} particle at every iteration t , where $V_i^t \geq 0$, as seen in equation 4.3. It is a three-part equation, where in the first part is the inertia factor $\omega^t V_i^t$, whose value is calculated from equation 4.4 in every iteration. A balance between the global and local search is established by assigning a positive linearly varying value. It should start with a global search then adopt a local search later in the process, which means that its value is ω_{max} at the beginning then linearly decreases to ω_{min} . The second term refers to updating the position based on its individual best, and the third represents the sharing of information of the global best to the rest of the positions.

Another equation 4.5 is added as an acceleration factor, to allow the global optimum to guide the individual optimization process. The equations are:

$$V_i^{t+1} = \omega^t V_i^{t+1} + c_1 r_1 (P_{best} - X_i^t) + c_2 r_2 (G_{best} - X_i^t) \quad (4.4)$$

$$\omega^t = \omega_{max} + \frac{\omega_{max} - \omega_{min}}{t_{max}} \times t \quad (4.5)$$

$$V_i^{t+1} = V_i^{t+1} + c_3 r_3 (G_{best}^t + P_{best}^t - 2X_i^t) + c_4 r_4 (G_{best}^t - P_{best}^t) \quad (4.6)$$

where, ω is the inertia factor, c_1, c_2, c_3, c_4 are constant values, r_1, r_2, r_3, r_4 are random values between 0 and 1, i is the current particle and t is the current iteration, t_{max} is the maximum number of iterations.

4.2.1.2 Velocity Limit Control

The velocity step length is limited according to the maximum/minimum values of the position, which is suitable when moving to the achieved optimum solution. The limits are calculated as per equation 4.7.

$$V_i^{limits} = \pm\beta(X_i^{max} - X_i^{min}) \quad (4.7)$$

where, β is the velocity limit operator.

4.2.1.3 Updating Particle Position

The particle transfers towards the optimum updating its location based on the velocity step function, which is added to the present position. It is updated based on equation 4.8.

$$X_i^{t+1} = X_i^t + V_i^{t+1} \quad (4.8)$$

4.2.1.4 Search space limit reduction strategy

The search area of each individual particle is decreased in each iteration as per the global optimum value. This involves adjusting the maximum and minimum values of the positions limits as per the equations in 4.9 and 4.10.

$$X_{imax}^{t+1} = X_{imax}^t - \sigma(X_{imax}^t - G_{best}^t) \quad (4.9)$$

$$X_{imin}^{t+1} = X_{imin}^t + \sigma(G_{best}^t - X_{imin}^t) \quad (4.10)$$

where, σ is the limit reduction strategy operator.

4.2.1.5 Resetting of Particle System

In order for the controller to account for changes in the solar irradiation, the algorithm is reset if the criteria shown in equation 4.11 is satisfied.

$$\frac{P_{prev} - P_{current}}{P_{prev}} \geq 10\% \quad (4.11)$$

The flowchart for the algorithm is shown in figure 4.3.

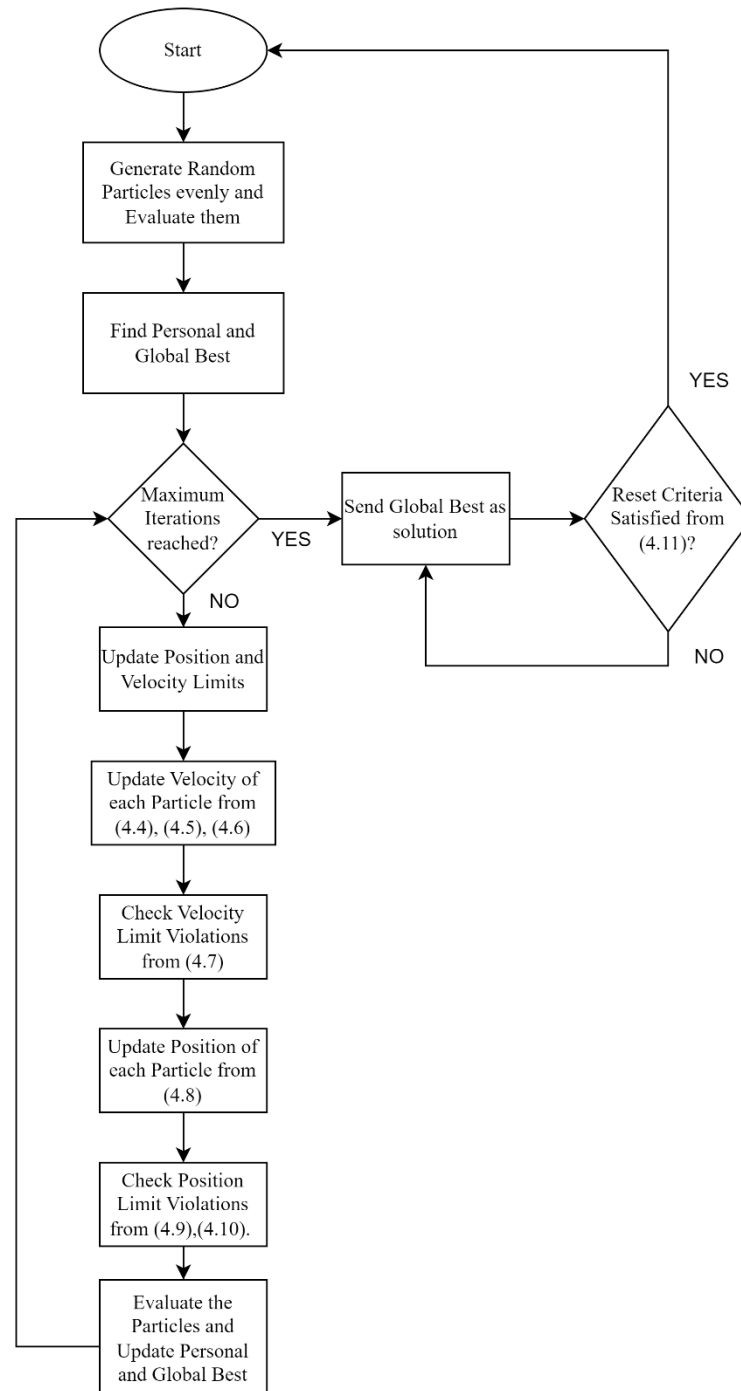


Fig. 4.3 Flowchart of the proposed MPSO

Chapter 5

MATLAB/Simulink Simulation

5.1 Simulink Models

The Simulink diagram of the PV system with a boost converter, and MPPT algorithms are shown in figures 5.1, 5.2. Figure 5.1 shows the PV module configuration used for simulation; it is a series connection of four PV modules with a bypass diode connected across each of them. Figure 5.2 shows the implementation of the block diagram of figure 3.12.

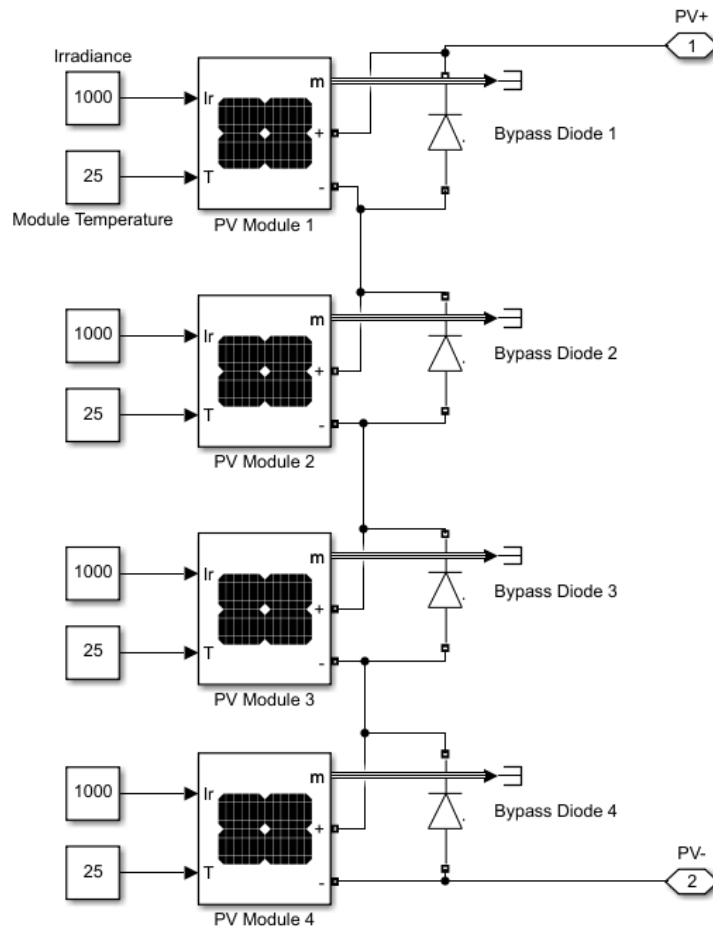


Fig. 5.1 Series connection of four PV modules

The MATLAB Script file used to implement the MPSO MPPT technique is given in Appendix-I.

5.2 Model Parameters

The parameters of the PV system, DC-DC boost converter, P&O algorithm, and the MPSO algorithm are list in Tables 5.1-5.4.

Table 5.1. Photovoltaic Module Specifications	
Parameters	Values
Maximum Power, P_m (W)	250
Maximum Power Voltage, V_m (V)	30.2
Maximum Power Current, I_m (A)	8.28
Open-Circuit Voltage, V_{oc} (V)	36
Short-Circuit Voltage, I_{sc} (A)	8.56
Number of cells per module, N_{cell}	60
Temperature coefficient of V_{oc} (%/C°)	-0.36099
Temperature coefficient of I_{sc} (%/C°)	0.102

Table 5.2. DC-DC Boost Converter Specifications	
Parameters	Values
Inductance, L (mH)	1.1478
Input Capacitance, C_i (mF)	1.75
Output Capacitance, C_o (mF)	0.4676
Load Resistance, R_L (Ω)	100
Switching Frequency, f_s (kHz)	50

Table 5.3. P&O Algorithm Specifications	
Parameters	Values
Step, dD	0.01
Initial Duty	0.5
Maximum Duty	1
Minimum Duty	0

Table 5.4. MPSO Algorithm Specifications	
Parameters	Values
Population, P	10
Iterations, t	10
Maximum Inertia, ω_{max}	1
Minimum Inertia, ω_{min}	0.4
Constants (c_1, c_2, c_3, c_4)	2.05, 2.05, 10, 0.0025
Velocity Limit Operator, β	0.1
LRS Operator, σ	0.1

5.3 Irradiance Data Inputs

The MPPT algorithms are tested for uniform and non-uniform irradiances at a constant temperature of 25°C, with their associated maximum power, voltage, and current generated by the PV array. The combinations used are listed in Table 5.4. The table also lists the values of the power for each of the peaks present in the P-V characteristic curves.

Table 5.5. Irradiance Data Inputs								
Cases	Radiations (W/m ²)				Peak 1	Peak 2	Peak 3	Peak 4
	PV 1	PV 2	PV 3	PV 4	(W)	(W)	(W)	(W)
Case 1	1000	1000	1000	1000	990.87	-	-	-
Case 2	1000	900	500	1000	482.28	693.98	555.13	-
Case 3	1000	400	600	400	227.95	308.84	420.66	-
Case 4	900	1000	1200	900	273.79	504.49	931.44	-
Case 5	500	800	1200	650	273.68	411.35	519.61	548.1
Case 6	700	1000	650	900	227.34	448.24	556.27	695.04

The P-V characteristic curves for each of the individual cases is shown from figures 5.4 to 5.10.

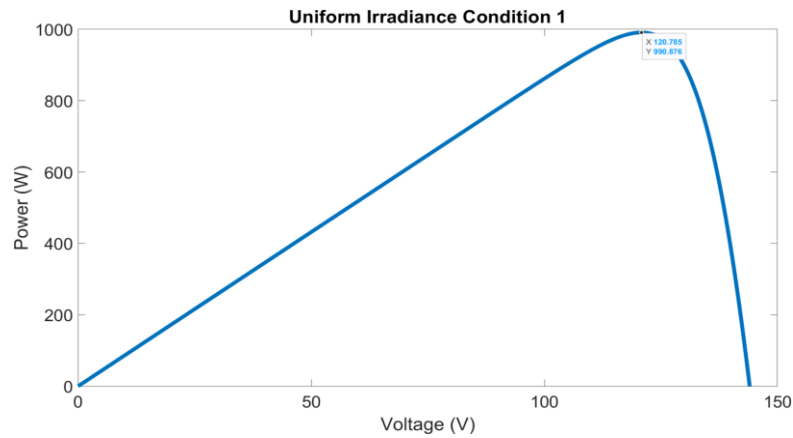


Fig. 5.4 P-V curve for Case-1

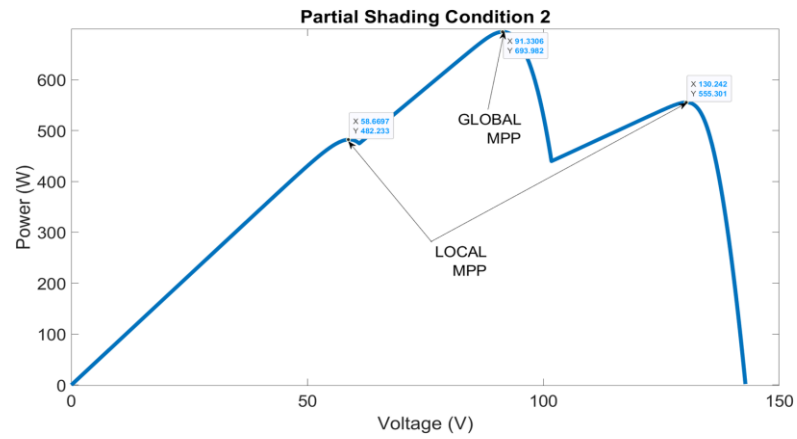


Fig. 5.5 P-V curve for Case-2

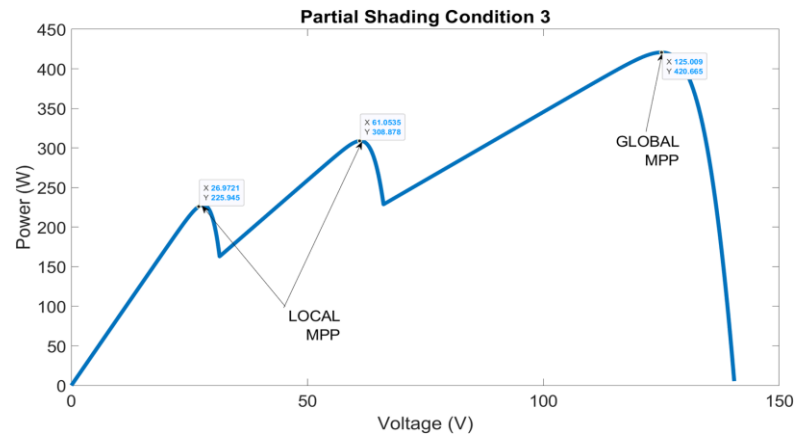


Fig. 5.6 P-V curve for Case-3

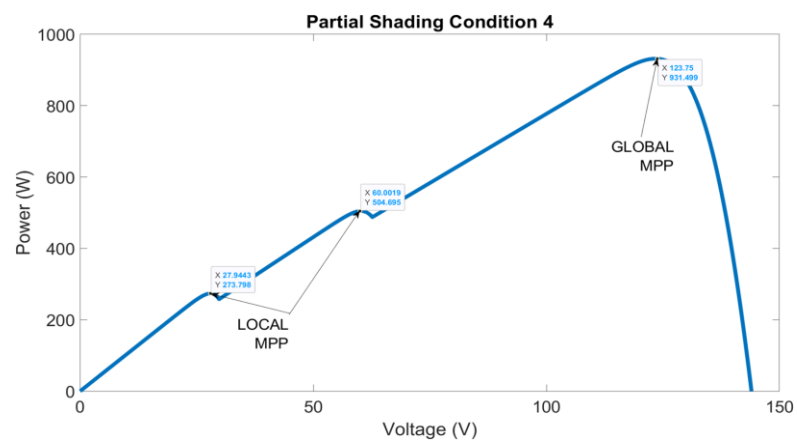


Fig. 5.7 P-V curve for Case-4

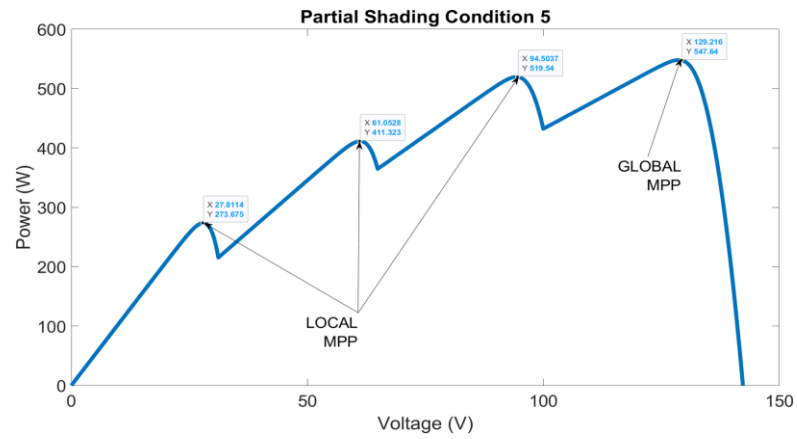


Fig. 5.8 P-V curve for Case-5

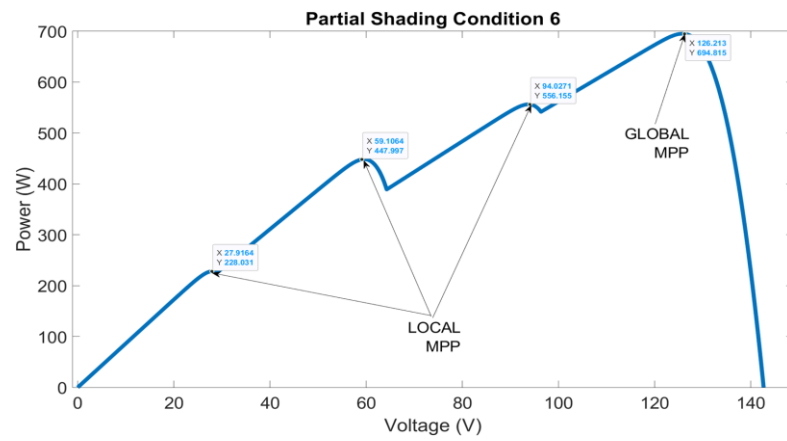


Fig. 5.9 P-V curve for Case-6

Chapter 6

Results and Discussions

The waveforms results shown in this section show four quantities in two separate windows, the first window shows the correct power to be tracked, and the PV power tracked by the algorithms. The second window shows the correct voltage to be tracked, and the PV voltage tracked by the algorithms.

Figures 6.1 and 6.2 show the power and voltage waveform results for Case 1 (1000, 1000, 1000, 1000 W/m²), both the algorithms accurately track the MPP of 990.87W, with the MPSO algorithm tracking 975.5W.

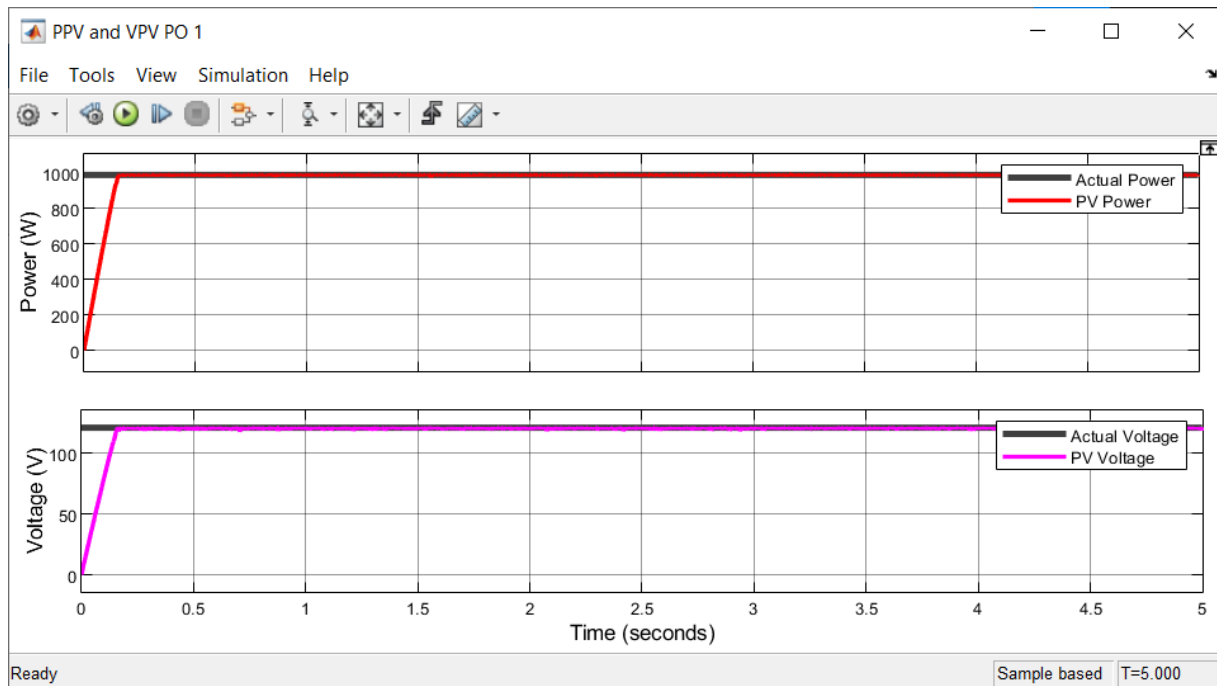


Fig. 6.1 Comparison of Actual Power and Voltage with values from P&O method for Case-1

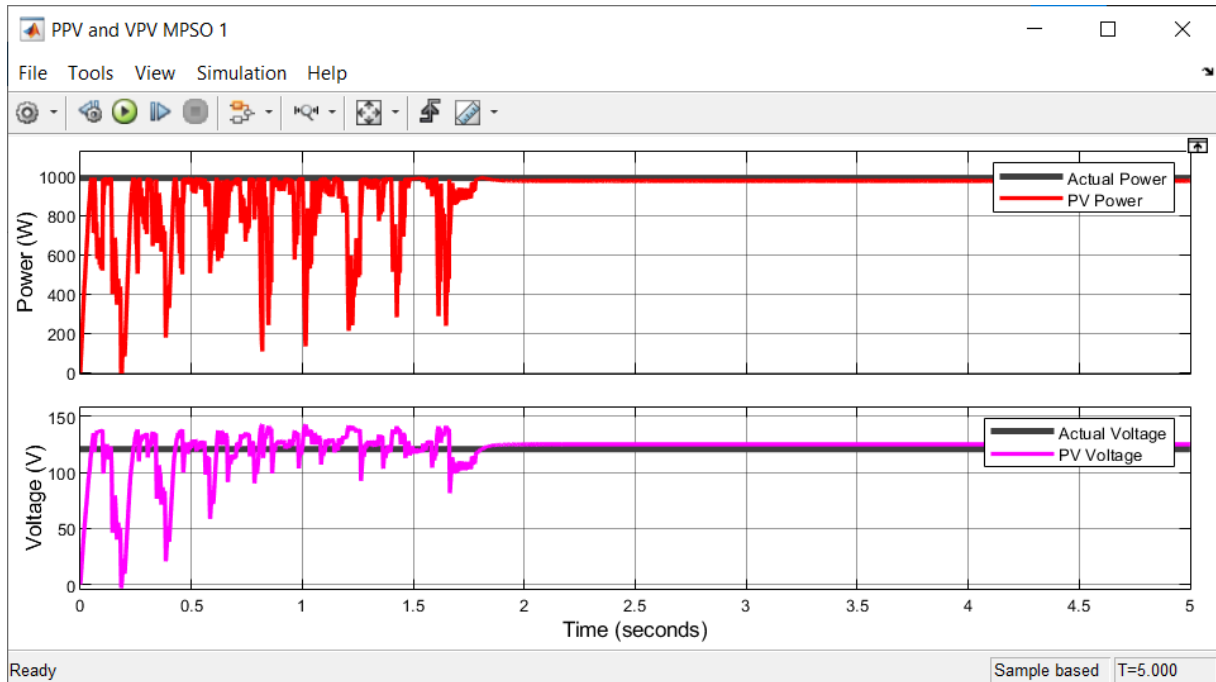


Fig. 6.2 Comparison of Actual Power and Voltage with values from MPSO method for Case-1

Figures 6.3 and 6.4 show the power and voltage waveform results for Case 2 (1000, 900, 500, 1000 W/m²), the P&O algorithm fails to track the 693.98W GMPP and outputs 482.2W, whereas the MPSO algorithm succeeds in tracking with an output of 692.4W.

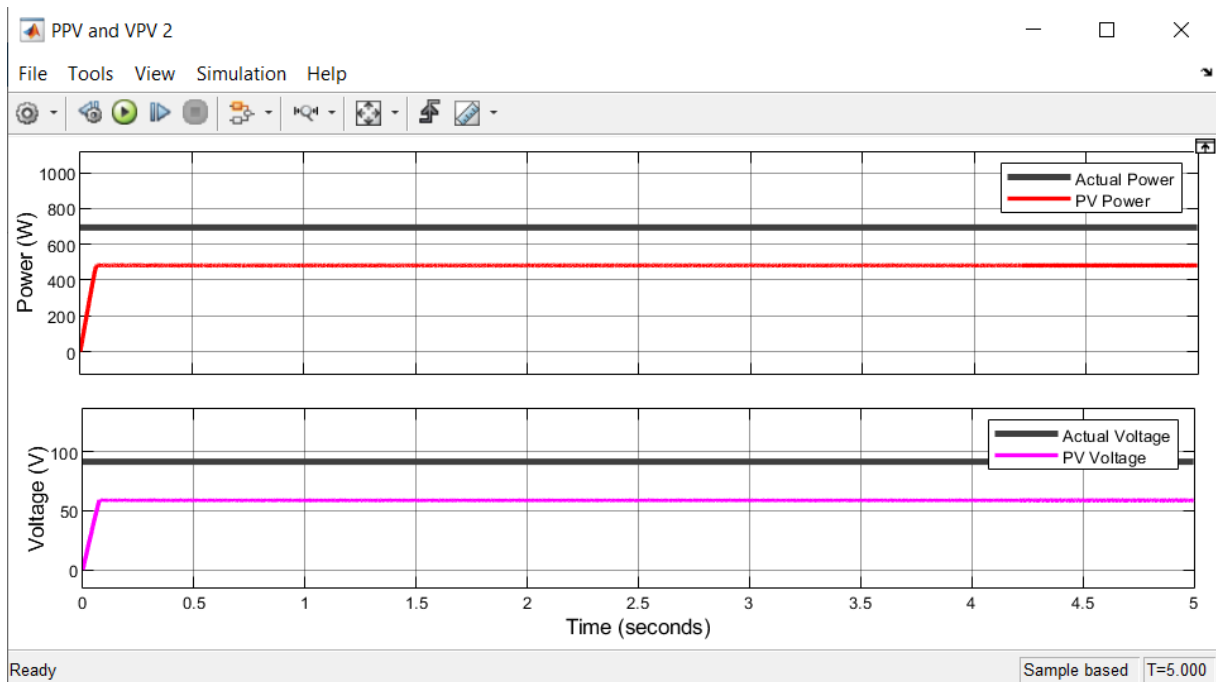


Fig. 6.3 Comparison of Actual Power and Voltage with values from P&O method for Case-2

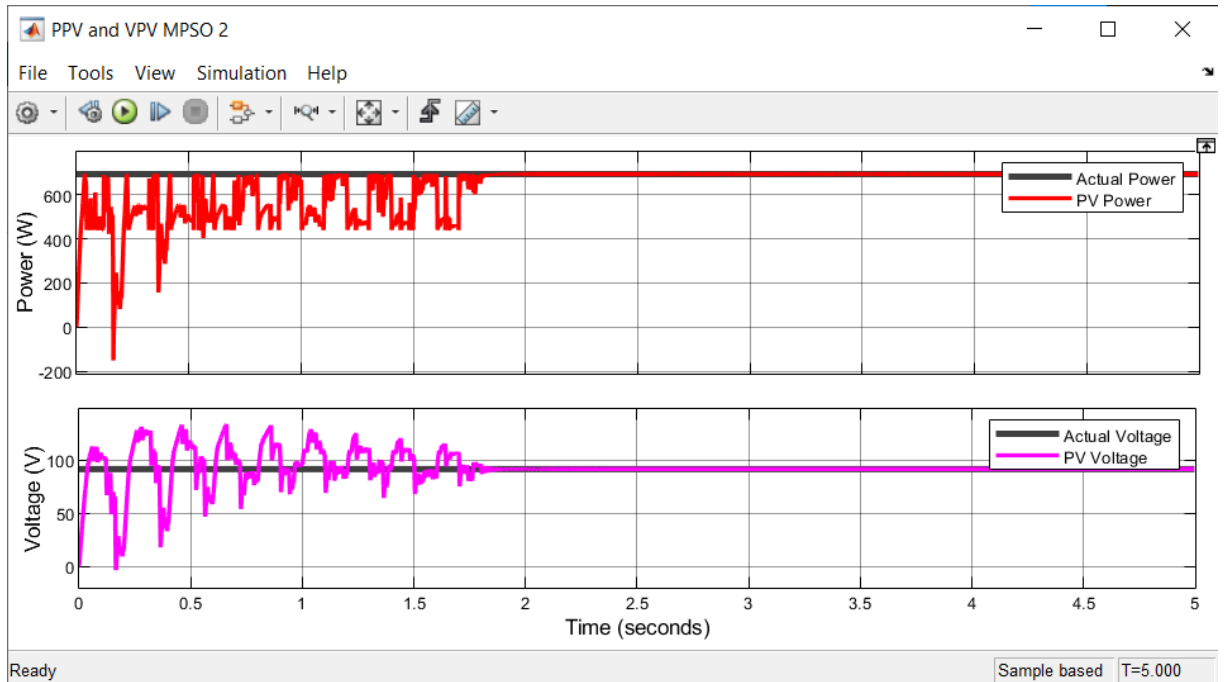


Fig. 6.4 Comparison of Actual Power and Voltage with values from MPSO method for Case-2

Figures 6.5 and 6.6 show the power and voltage waveform results for Case 3 (1000, 400, 600, 400 W/m²), the P&O algorithm fails to track the 420.66W GMPP and outputs 227.8W, whereas the MPSO algorithm succeeds in tracking with an output of 412W.

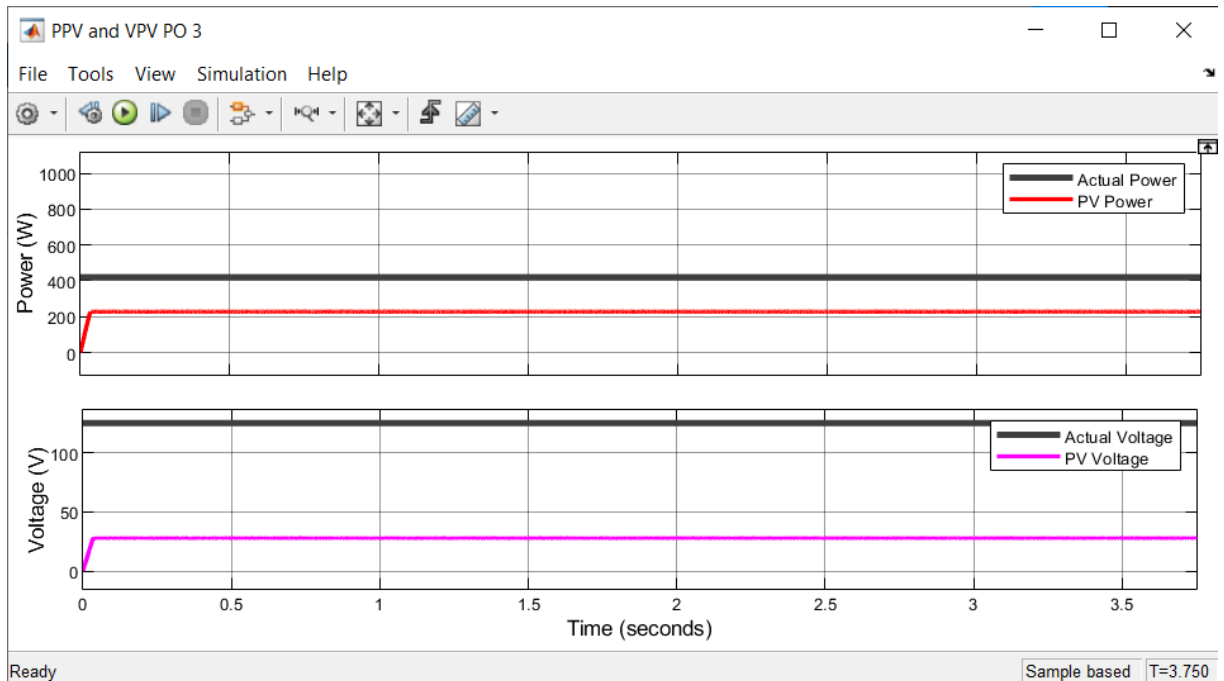


Fig. 6.5 Comparison of Actual Power and Voltage with values from P&O method for Case-3

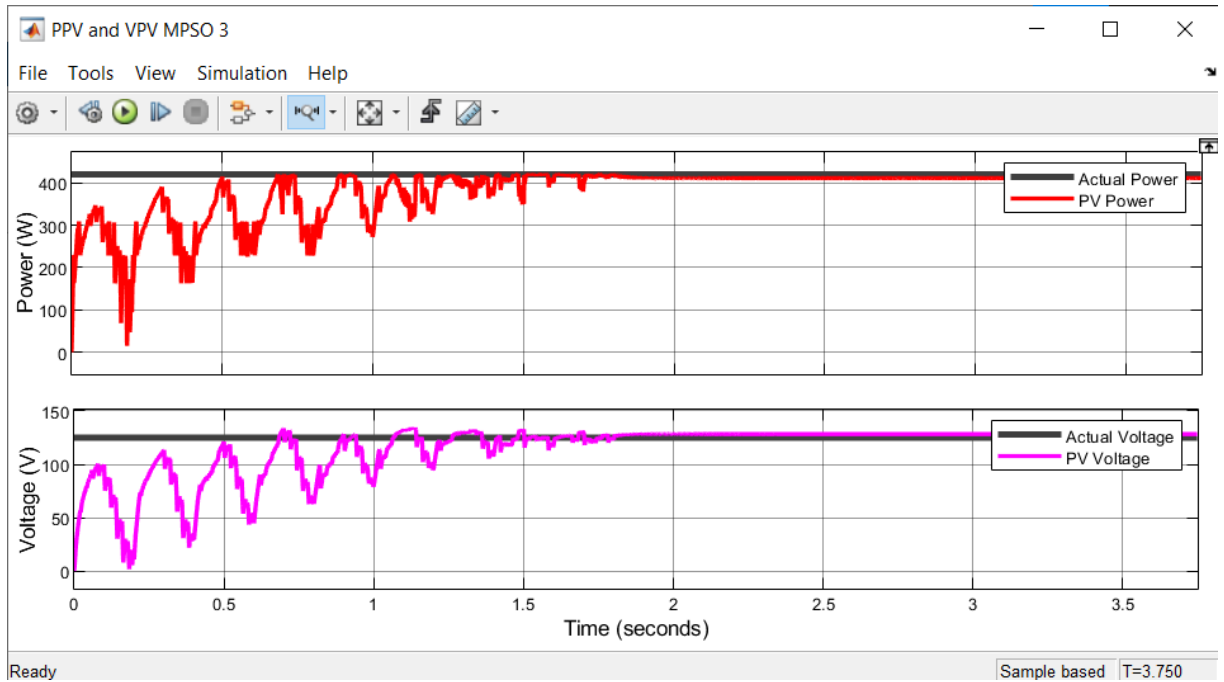


Fig. 6.6 Comparison of Actual Power and Voltage with values from MPSO method for Case-3

Figures 6.7 and 6.8 show the power and voltage waveform results for Case 4 (1000, 400, 600, 400 W/m²), the P&O algorithm fails to track the 931.44W GMPP and outputs 273.5W, whereas the MPSO algorithm succeeds in tracking with an output of 925.3W.

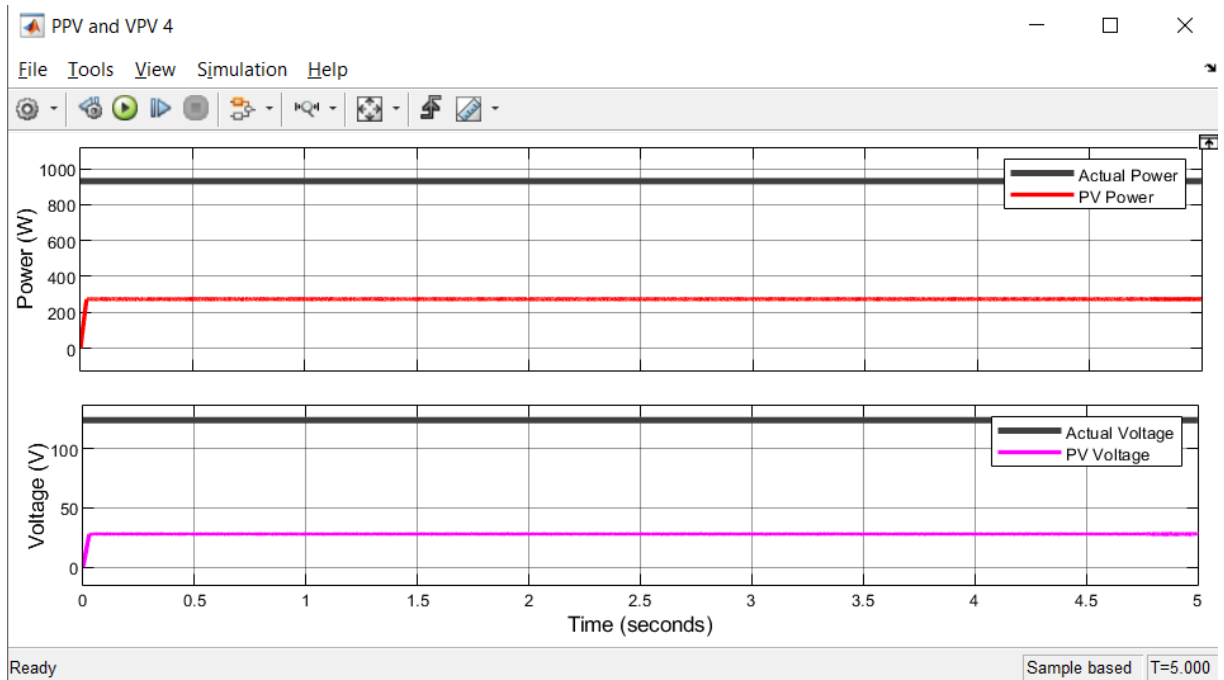


Fig. 6.7 Comparison of Actual Power and Voltage with values from P&O method for Case-4

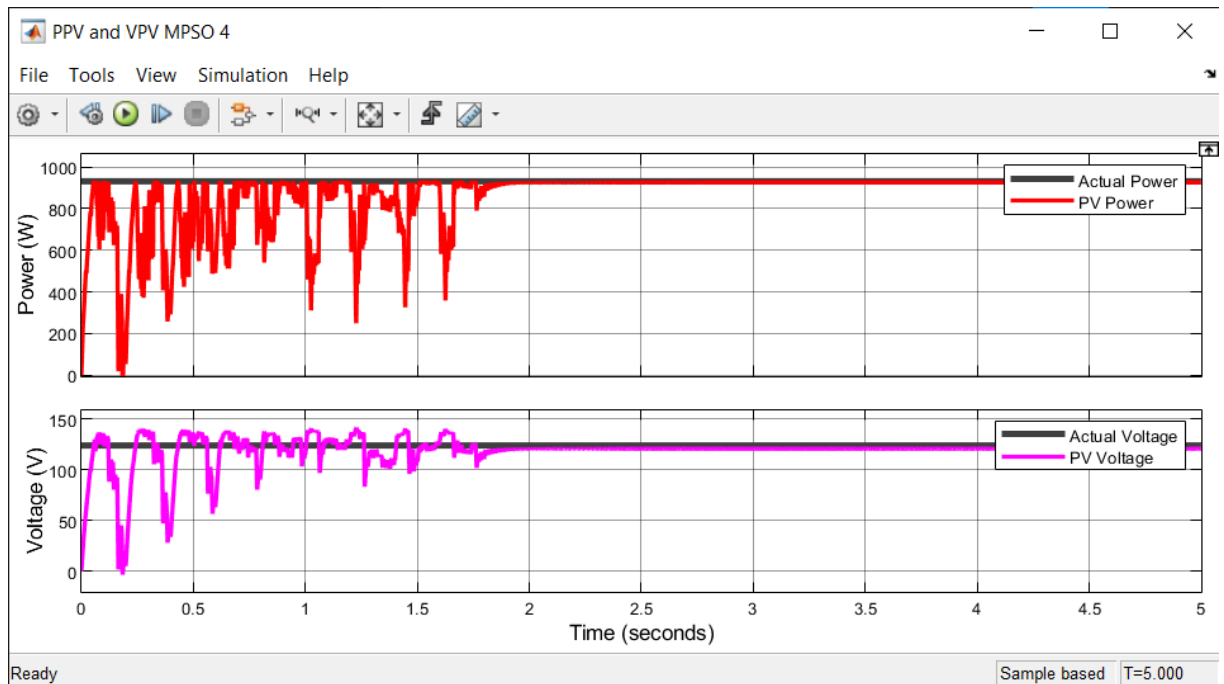


Fig. 6.8 Comparison of Actual Power and Voltage with values from MPSO method for Case-4

Figures 6.9 and 6.10 show the power and voltage waveform results for Case 5 (500, 800, 1200, 650 W/m²), the P&O algorithm fails to track the 548.1W GMPP and outputs 273.5W, whereas the MPSO algorithm succeeds in tracking with an output of 541.5W.

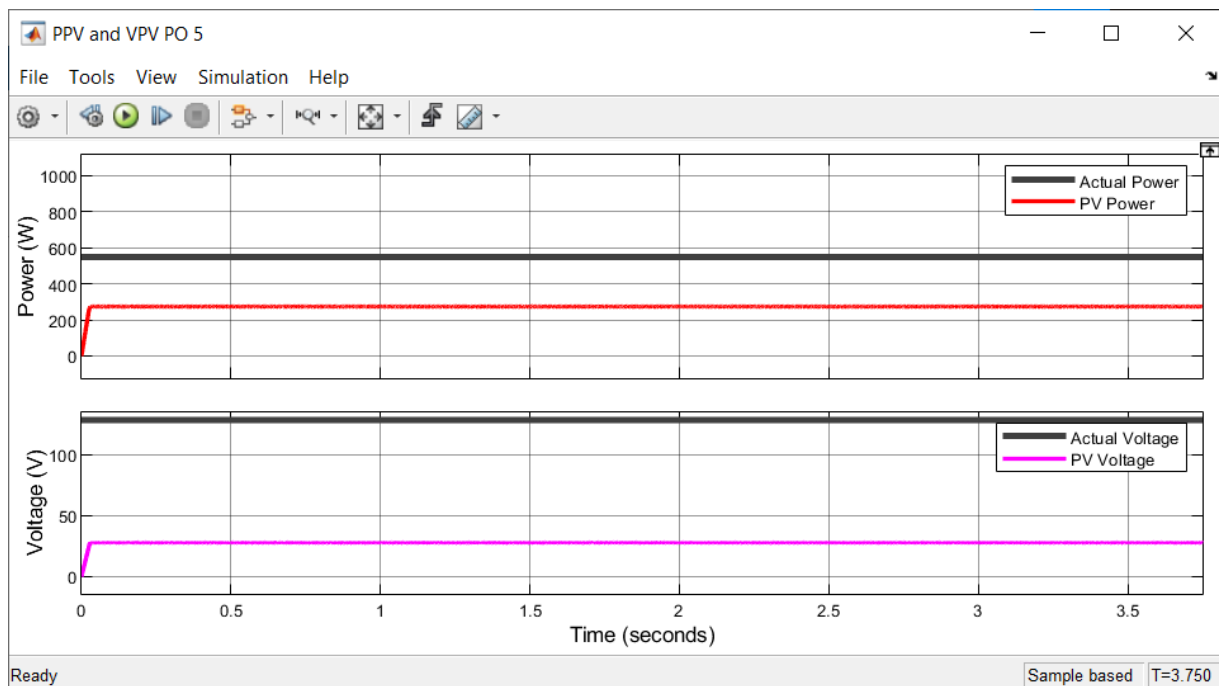


Fig. 6.9 Comparison of Actual Power and Voltage with values from P&O method for Case-5

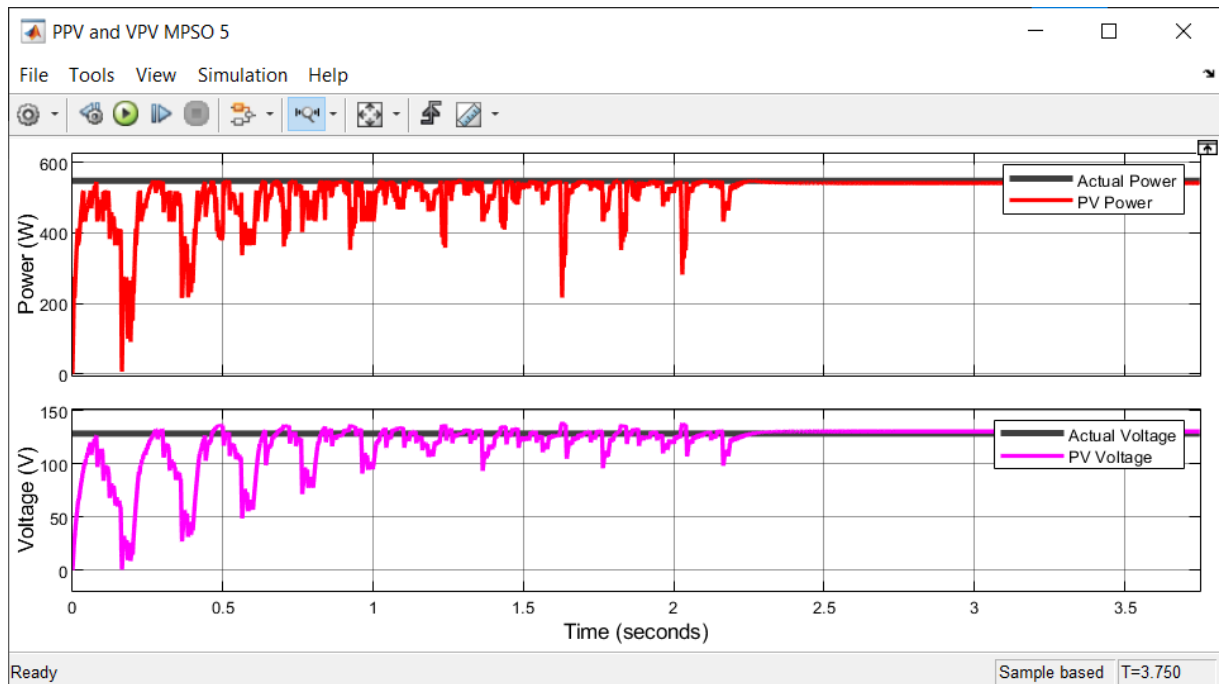


Fig. 6.10 Comparison of Actual Power and Voltage with values from MPSO method for Case-5

Figures 6.11 and 6.12 show power and voltage the waveform results for Case 6 (700, 1000, 650, 900 W/m²), the P&O algorithm fails to track the 695.04W GMPP and outputs 227.9W, whereas the MPSO algorithm succeeds in tracking with an output of 693.9W.

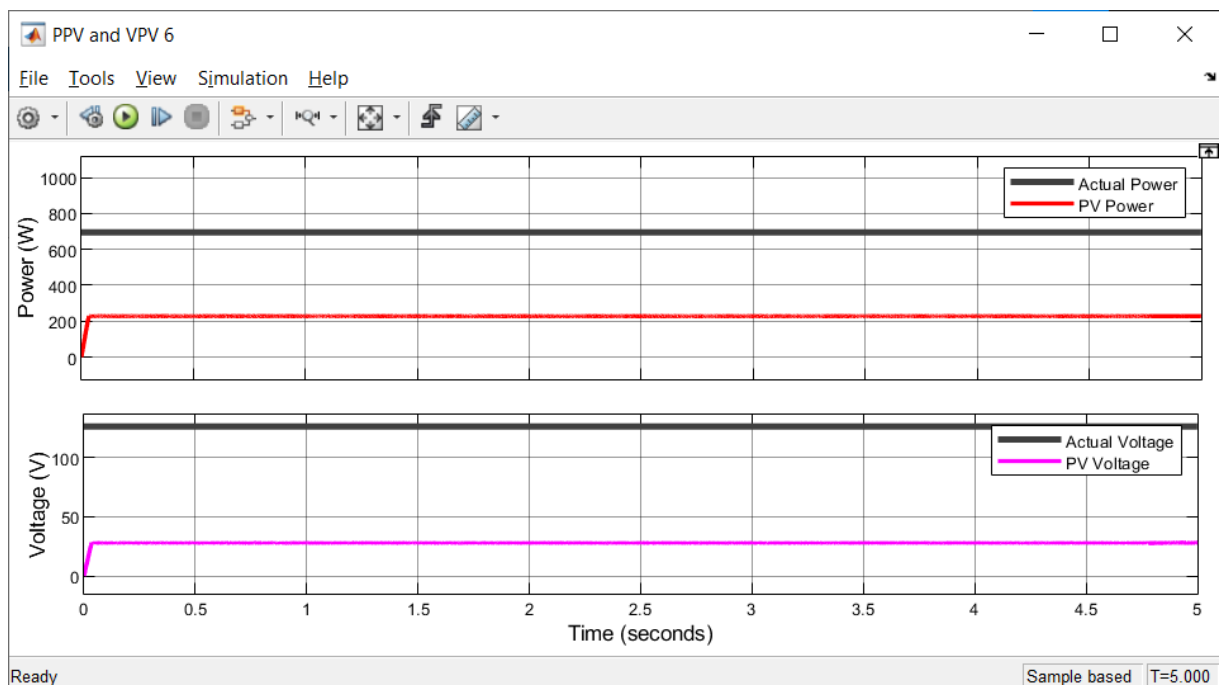


Fig. 6.11 Comparison of Actual Power and Voltage with values from MPSO method for Case-6

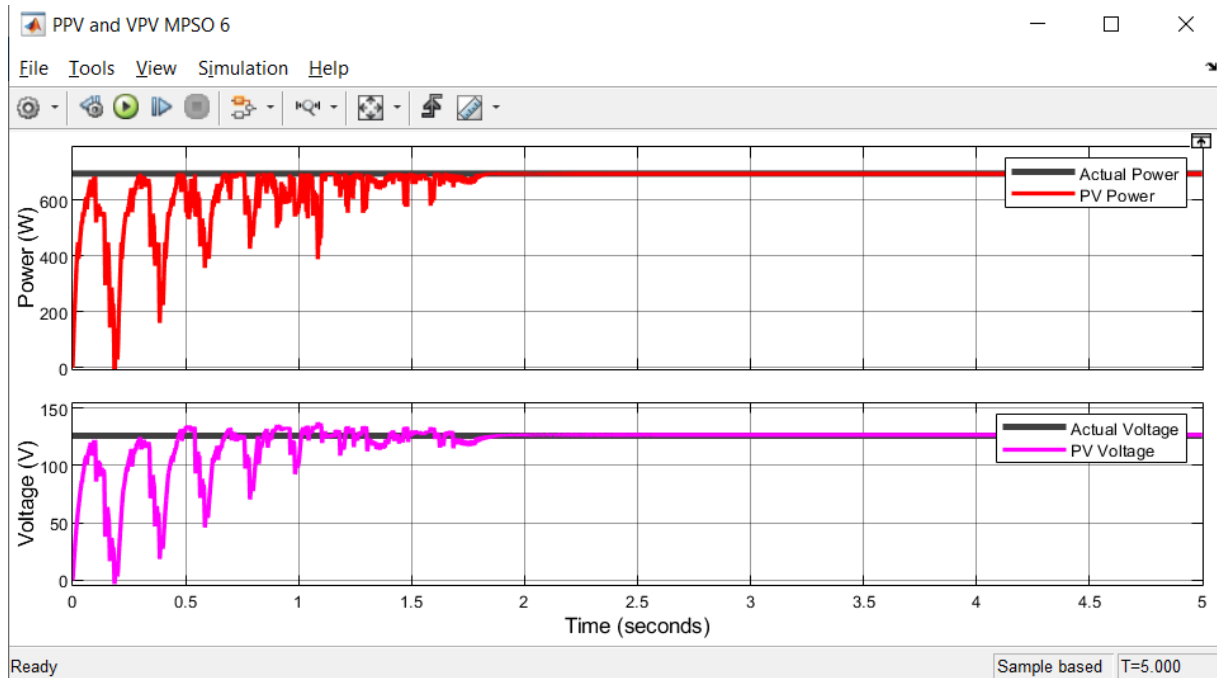


Fig. 6.12 Comparison of Actual Power and Voltage with values from MPSO method for Case-6

Figure 6.13 shows the performance of the algorithm under dynamic PSC as the radiation profile across the PV configuration changes from Case 1 (1000, 1000, 1000, 1000 W/m²) to Case 4 (1000, 400, 600, 400 W/m²) at simulation time = 5s for a total simulation time = 10s.

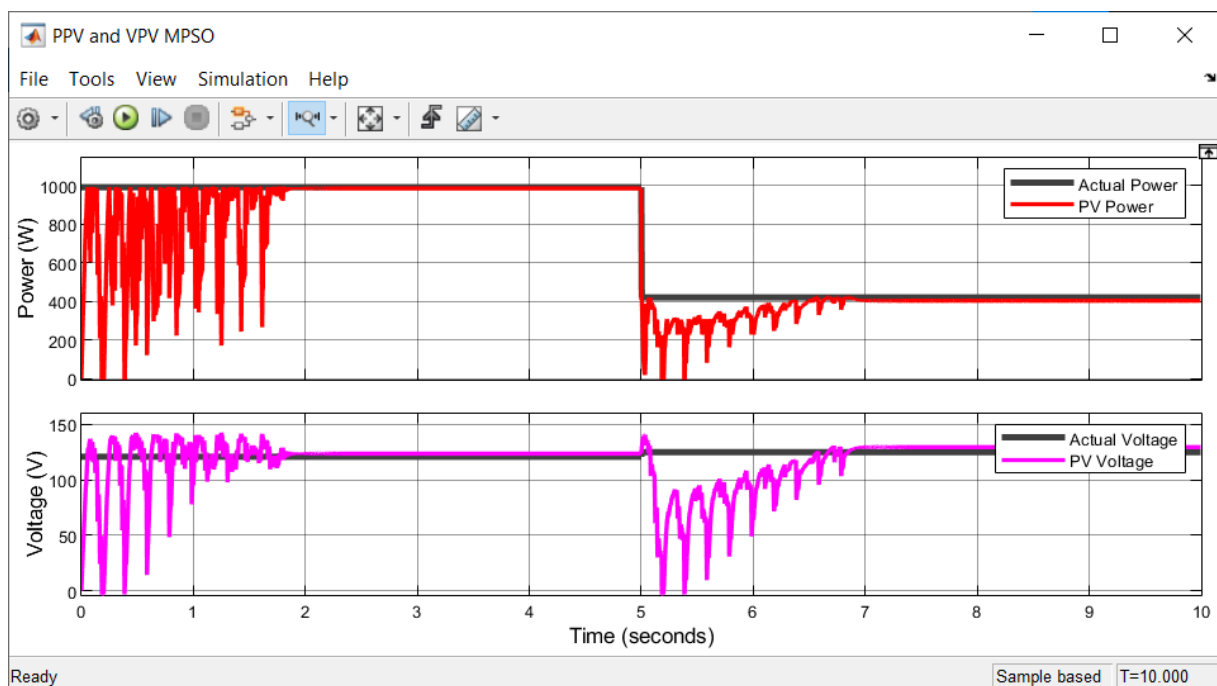


Fig. 6.13 Comparison of Actual Power and Voltage with values from MPSO method as irradiance changes from Case-1 to Case-4

The efficiency is an important principle that is utilized to assess the performance of MPPT algorithms and is given by equation 6.1 as shown:

$$\eta\% = \left(1 - \frac{P_{\max-Actual} - P_{\max-Method}}{P_{\max-Actual}}\right) \times 100 \quad (6.1)$$

The values obtained by the algorithms are given in Table 6.1. It can be seen that the P&O algorithm is only successful in tracking the power only in the case of uniform radiation with an efficiency of 99.969%. While for the other PSC cases, it fails to obtain the GMPP and tracks the first LMPP it comes across.

In case of the MPSO technique, regardless of the irradiation, the algorithm is able to track the GMPP with good stabilization. Moreover, the efficiencies of >95% was observed for all cases.

Table 6.1. Performance Efficiency for Algorithms					
Cases	P _{max-Actual} (W)	P _{max-P&O} (W)	Efficiency of P&O (%)	P _{max-MPSO} (W)	Efficiency of MPSO (%)
Case 1	990.87	990.6	99.969	975.5	98.475
Case 2	693.98	482.2	69.483	692.4	99.772
Case 3	420.66	227.8	54.152	412	97.941
Case 4	931.44	273.5	29.363	925.3	99.340
Case 5	548.1	273.5	49.899	541.5	98.795
Case 6	695.04	227.9	32.789	693.9	99.835

Chapter 7

Conclusions and Future Scope

7.1 Conclusions

An important objective of MPPT methods is that they are primarily used to obtain the GMPP of PV systems to provide this power to the consumer load for maximum efficiency. Under uniform irradiance, the MPP can be easily obtained using any simple control technique such as those in the conventional category. But under PSC, this extraction of GMPP becomes difficult as multiple LMPPs also get generated which simple methods fail to extract.

In this report, a modified PSO technique is introduced and its performance was evaluated and compared with that of the simple P&O method, for six difference cases of irradiation profiles. The P&O failed to track the GMPP for all the PSC cases and succeeded only for the first uniform radiation case. The modified PSO was shown to track the GMPP effectively and robustly as required for all the cases. It also is very easy to implement. It has an excellent tracking performance in terms of its efficiency, as shown by the simulation results.

Despite its high tracking efficiency, one major drawback with this algorithm is its relative slow tracking speed as compared to the P&O algorithm, making it unsuitable for tracking the GMPP under conditions with rapidly changing irradiation profiles.

7.2 Future Scope

Algorithms that are developed in the future should be aimed at improving the convergence speed, tracking accuracy of the obtained GMPP. Moreover, they should be easy to implement and relatively cheap.

One way to improve these methods is to adopt hybrid MPPT techniques which integrate AI-based techniques with that of conventional techniques, such that it has the advantages of both whilst overcoming the disadvantages of the individual methods. For example, a hybrid PSO-P&O MPPT technique can be used in which, the PSO can be made to bring the operating point near the GMPP region, after which it transfers the control to the P&O algorithm. The P&O will control the tracking of the GMPP under varying irradiance conditions, and the PSO algorithm is reset whenever the drop in the output power exceeds a threshold value.

Appendix I

MPSO MATLAB Script File

```
function Duty = MPPT(Vpv,Ipv)

    persistent D;
    persistent V;
    persistent D_PB;
    persistent P_PB;
    persistent D_GB;
    persistent P_GB;
    persistent D_MAX;
    persistent D_MIN;
    persistent V_MAX;
    persistent V_MIN;
    persistent D_PREV;
    persistent W;
    persistent ITER;
    persistent P;
    persistent FLAG_ZERO;
    persistent FLAG_CONV;
    persistent FLAG_RESET;
    persistent DELAY;
    persistent AVG;
    persistent Ppv;
    persistent Pprev;

    if isempty(FLAG_RESET) && isempty(Ppv)
        FLAG_RESET = 0;
        Ppv = 0;
    end

    l_limit = 0;
    u_limit = 1;
    C1 = 2.05;
    C2 = 2.05;
    C3 = 10;
    C4 = 0.0025;
    beta = 0.1;
```

```

sigma = 0.1;
wmax = 1;
wmin = 0.4;

T_MAX = 10; % Maximum Iterations
M = 10; % Number of Particles
SET = 2000;
PET = SET * 0.5; %0.25

if(isempty(D) || FLAG_RESET == 1)
    rng('shuffle');
    D = divideDuty(M,u_limit,l_limit)
    D_PREV = D;
    V = zeros(M,1);
    D_PB = D
    D_GB = 0; % Maximizing Objective Function
    P_PB = zeros(M,1);
    P_GB = 0;

    D_MIN = zeros(M,1);
    D_MAX = zeros(M,1);
    D_MAX(:, :) = 1;
    V_MIN = zeros(M,1);
    V_MIN = -1 * beta * (D_MAX - D_MIN);
    V_MAX = zeros(M,1);
    V_MAX = beta * (D_MAX - D_MIN);

    W = zeros(M,1);
    W(:, :) = wmax;
    ITER = 0; % Initial Iteration
    P = 1; % Initial Particle

    FLAG_ZERO = 0;
    FLAG_CONV = 0;
    FLAG_RESET = 0;
    DELAY = 1;
    AVG = 0;
    Duty = D(P); % Initial Duty Cycle P = 1
    P = P + 1;
    return;
end

% Delay for settling
if DELAY <= SET && FLAG_CONV == 0
    if DELAY >= PET % Averages the Power obtained
        AVG = AVG + (Vpv * Ipv);
    end
    Duty = D(P-1);
    DELAY = DELAY + 1;
    return;
end

```

```

if FLAG_CONV == 1 % Checking for Reset Criteria
    Pprev = Ppv
    Ppv = Vpv * Ipv
    abs((Ppv - Pprev) / Pprev)
    if abs((Ppv - Pprev) / Pprev) > 0.1
        FLAG_RESET = 1;
        disp("Reset Flag Raised");
    end
    Duty = D_GB;
    return;
end

Pprev = Ppv;
Ppv = AVG / (SET - PET); % Objective Function
AVG = 0;

if ITER == 0 % Zero Pass
    P_PB(P-1) = Ppv;
% Personal Best Power, previous particle duty cycle evaluated
    if Ppv > P_GB
        D_GB = D(P-1);
        P_GB = Ppv;
    end
    if P == M + 1
        disp("Zero Flag Raised");
        FLAG_ZERO = 1;
        P = 1;
    else
        Duty = D(P);
        P = P + 1;
        DELAY = 1;
        return;
    end
end

end

if FLAG_CONV == 0 && ITER <= T_MAX
    if FLAG_ZERO == 0
        if Ppv > P_PB(P-1)
            D_PB(P-1) = D(P-1);
            P_PB(P-1) = Ppv;
        end
        if Ppv > P_GB
            D_GB = D(P-1);
            P_GB = Ppv;
        end
    end
end

```

```

        if P == M + 1 || FLAG_ZERO == 1
% Updating Positions and Velocities

% Limits Updation
D_MAX = D_MAX - sigma * (D_MAX - D_GB);
D_MIN = D_MIN + sigma * (D_GB - D_MIN);
V_MIN = -1 * beta * (D_MAX - D_MIN);
V_MAX = beta * (D_MAX - D_MIN);

% Velocity
R1 = rand(M,1);
R2 = rand(M,1);
R3 = rand(M,1);
R4 = rand(M,1);
V = (W .* V) + (C1 * (R1 .* (D_PB - D))) + (C2 *
(R2 .* (D_GB - D)));
W_temp = wmax - ((wmax - wmin) / T_MAX) * ITER;
W(:, :) = W_temp;
V = V + (C3 * (R3 .* (D_GB + D_PB - 2 * D))) + (C4
* (R4 .* (D_GB - D_PB)));

% Velocity Limits
idx_min = find(V < V_MIN);
V(idx_min) = V_MIN(idx_min);
idx_max = find(V > V_MAX);
V(idx_max) = V_MAX(idx_max);

% Position Update
D_PREV = D;
D = D + V;

% Position Limits
idx_min = find(D < D_MIN);
D(idx_min) = D_MIN(idx_min);
idx_max = find(D > D_MAX);
D(idx_max) = D_MAX(idx_max);

if (abs((D_PREV - D) ./ D_PREV) < 0.01)
    disp("Convergence Flag Raised");
    Duty = D_GB;
    FLAG_CONV = 1;
    return;
elseif ITER >= T_MAX
    disp("Iteration Flag Raised");
    Duty = D_GB;
    FLAG_CONV = 1;
    return;
end

```

```

        ITER = ITER + 1;
        P = 1;
        FLAG_ZERO = 0;

        % Restart Function Evaluation
        Duty = D(P);
        P = P + 1;
        DELAY = 1;
    else
        Duty = D(P);
        P = P + 1;
        DELAY = 1;
    end
else
    Duty = D_GB;
    DELAY = 1;
end

end

% This function randomly initializes the particles uniformly
across the 0 and 1 interval

function D = divideDuty(M,U,L)
    step = (U - L) / M;
    P = zeros(M,1);
    init_L = L;
    init_U = init_L + step;
    for i = 1:M
        P(i) = init_L + (init_U - init_L) * rand;
        init_L = init_U;
        init_U = init_L + step;
    end
    D = P;
end

```

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